LAB 6 :Dynamic Routing Configuration: RIP, EIPRG, OSPF, BGP

Objectives

- To understand how routers independently identify remote networks and maintain routing tables without requiring manual path configuration.
- To set up and evaluate the behavior of RIP, EIGRP, OSPF, and BGP, examining how each algorithm — Distance Vector, Link-State, and Path Vector — autonomously detects network topologies and fills in routing tables.

Theory

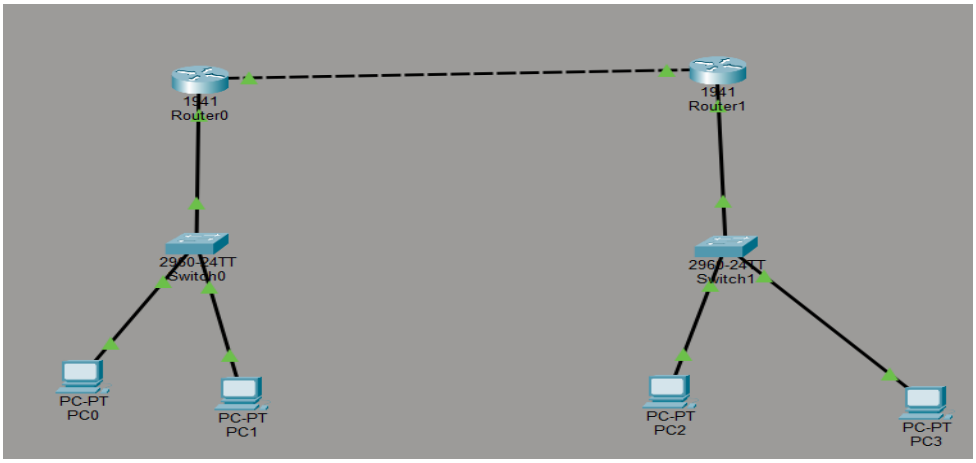
Dynamic routing protocols enable routers to autonomously identify remote networks and construct their own routing tables. In contrast to static routing, these protocols can respond to topology modifications without any manual input.

Routing Information Protocol (RIP)

RIP is among the earliest distance vector protocols. While it is straightforward to implement, it comes with notable constraints when deployed in larger networks.

- Algorithm: Employs the Distance Vector algorithm.
- Metric: Relies on Hop Count to determine the optimal path, with a ceiling of 15 hops.
- Operation: Each router broadcasts its complete routing table to adjacent routers every 30 seconds.
- Versions: RIP v1 operates in classful mode (no subnetting support), whereas RIP v2 is classless and is compatible with CIDR.

Network Topology



Configuration :

For PCs :

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1
PC2	192.168.2.2	255.255.255.0	192.168.2.1
PC3	192.168.2.3	255.255.255.0	192.168.2.1

For Routers :

Device	IPv4 Address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 1	192.168.2.0	255.255.255.0	10.0.0.2	gig 0/0 gig 0/1	10.0.0.1 192.168.1.1
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0 gig 0/1	10.0.0.2 192.168.2.1

Commands

Router 0 :	**Router 1 :**
router rip	router rip
version 2	version 2
network 192.168.1.0	network 192.168.2.0
network 10.0.0.0	network 10.0.0.0
no auto-summary	no auto-summary

Observation

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

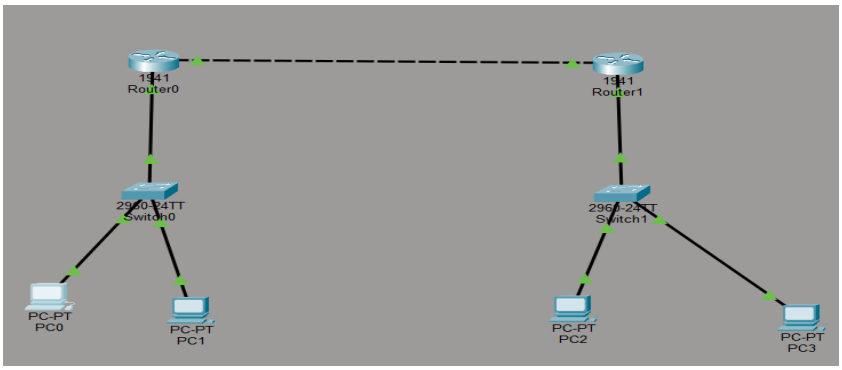
1. Enhanced Interior Gateway Routing Protocol (EIGRP)

EIGRP is a hybrid routing protocol that draws on the strengths of both distance vector and link-state approaches.

- Algorithm: Utilizes the Diffusing Update Algorithm (DUAL) to guarantee loop-free routing paths.
- Metric: Calculates routes using a composite metric derived from Bandwidth and Delay, with optional consideration of Reliability and Load.

- Operation: Route updates are only transmitted when a topology change is detected, thereby conserving bandwidth.
- Feature: Uniquely supports unequal cost load balancing — a capability not found in the other protocols covered here.

Network Topology



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For Routers :

Device	IPv4 Address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
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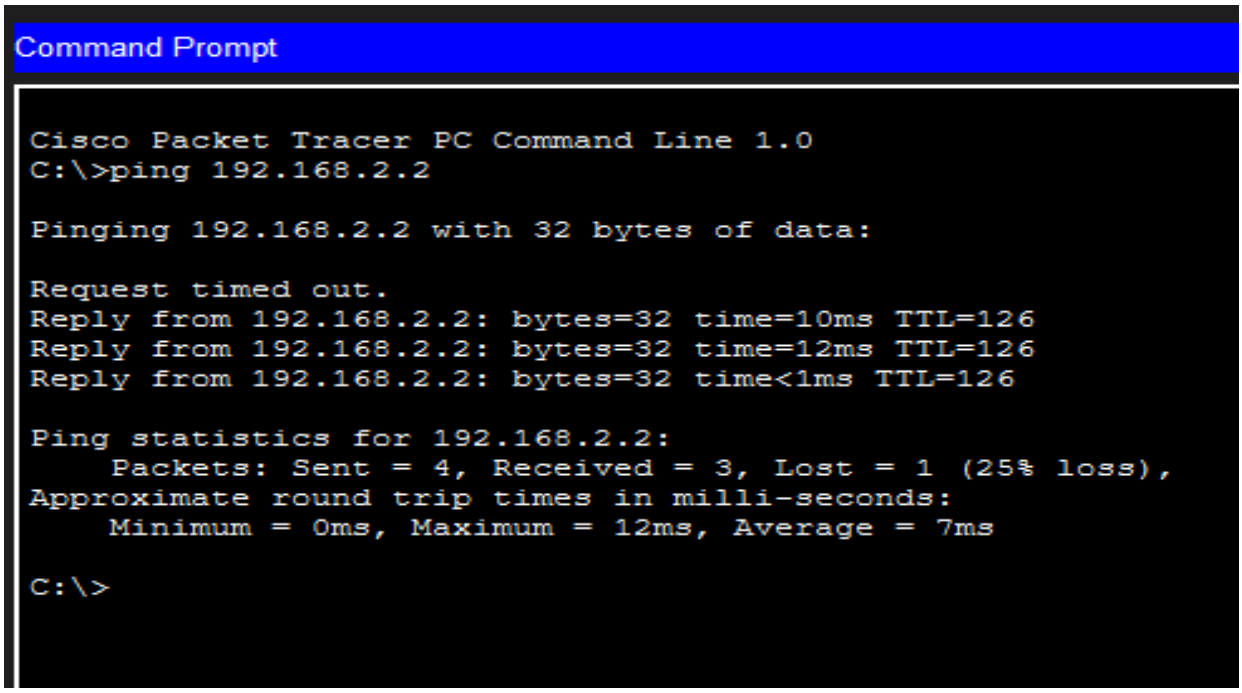
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0 gig 0/1	10.0.0.2 192.168.2.1

Commands

Router 0 :	**Router 1 :**
router eigrp 100	router eigrp 100
network 192.168.1.0 0.0.0.255	network 192.168.2.0 0.0.0.255
network 10.0.0.0 0.0.0.255	network 10.0.0.0 0.0.0.255
no auto-summary	no auto-summary

Observation

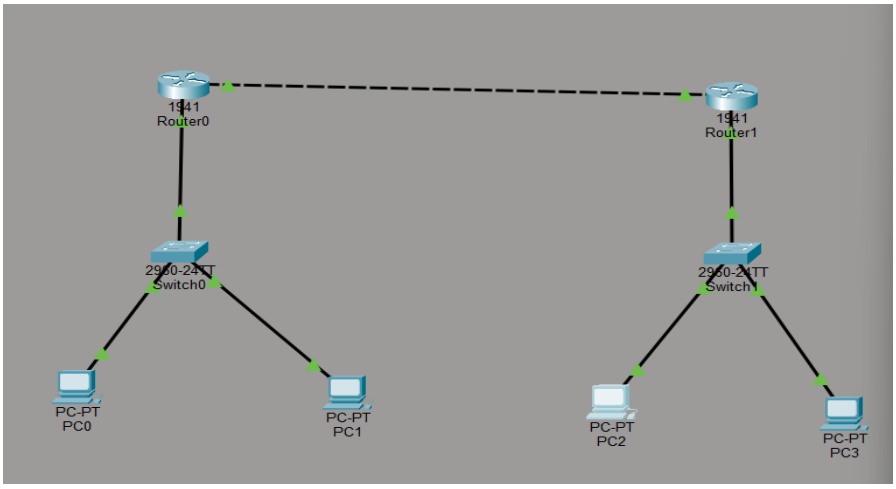


2. Open Shortest Path First (OSPF)

OSPF is a broadly adopted link-state routing protocol built for scalability and rapid network convergence.

- Algorithm: Applies Dijkstra's Shortest Path First (SPF) algorithm.
- Metric: Measures Cost, computed by dividing the reference bandwidth by the bandwidth of the interface.
- Operation: Neighbor discovery is achieved through "Hello" packets, after which routers exchange Link State Advertisements (LSAs) to establish a common topology database.
- Advantage: Delivers significantly faster convergence compared to RIP.

Network Topology



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Router 2	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0 gig 0/1	10.0.0.2 192.168.2.1

Commands

Router 0 :

router ospf 1

network 192.168.1.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.255 area 0

Router 1 :

router ospf 1

network 192.168.2.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.255 area 0

Observation

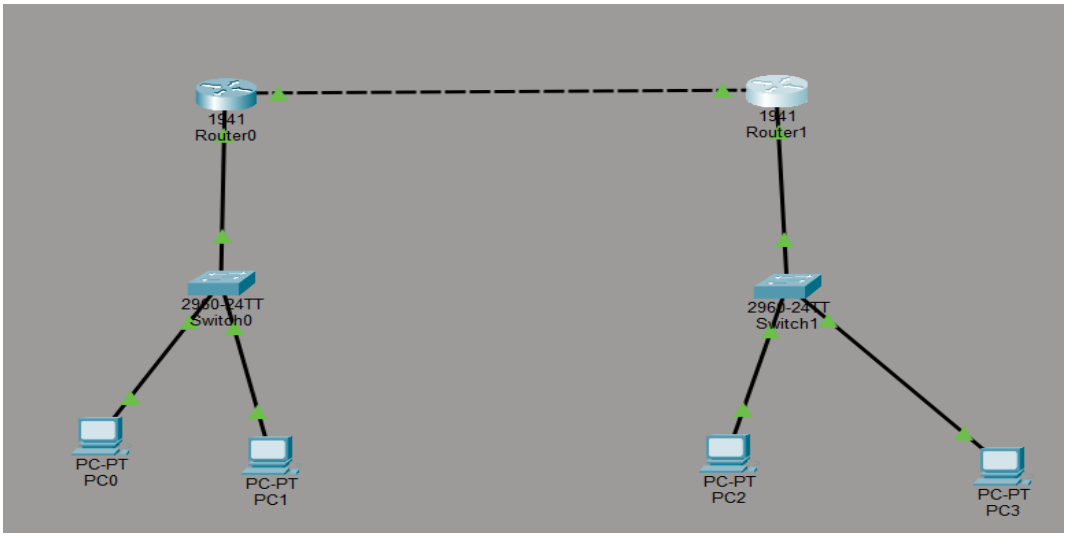
```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 182/168.2.2
Ping request could not find host 182/168.2.2. Please check t
C:\>192.168.2.2
Invalid Command.

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=3ms TTL=128
Reply from 192.168.2.2: bytes=32 time=4ms TTL=128
Reply from 192.168.2.2: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>
```

Border Gateway Protocol (BGP)

BGP is a path vector protocol widely regarded as the "routing protocol of the internet."

- Purpose: Designed to interconnect separate Autonomous Systems (AS) — distinct collections of networks managed under a single administrative domain.
- Operation: Rather than simple numeric metrics, BGP uses path attributes (such as AS-Path) to determine the preferred route.
- Reliability: Communication between routers is established reliably over TCP port 179.
- Scalability: Highly robust and capable of managing the entire global internet routing table.

Network Topology



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For Routers :

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Router 2	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0 gig 0/1	10.0.0.2 192.168.2.1

Commands

Assumption :

R1 → AS 65001 R2 → AS 65002

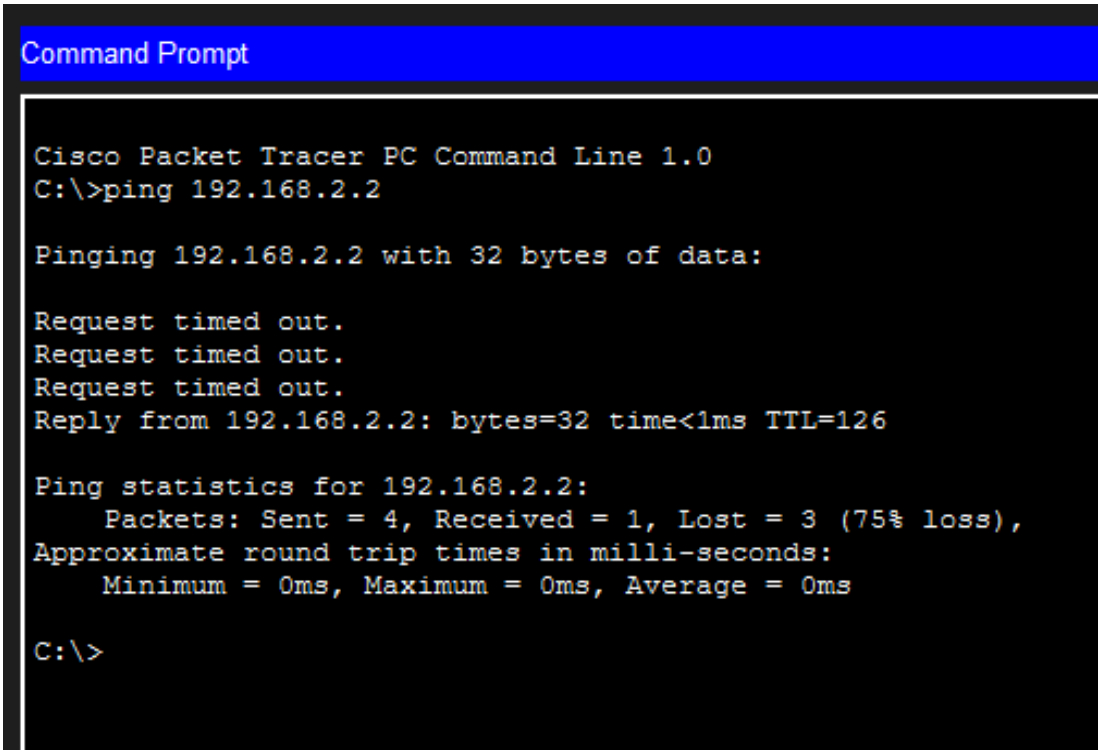
Router 0 :

```
router bgp 65001
neighbor 10.0.0.2 remote-as 65002
network 192.168.1.0 mask 255.255.255.0
```

Router 1 :

```
router bgp 65001
neighbor 10.0.0.1 remote-as 65001
network 192.168.2.0 mask 255.255.255.0
```

Observation



Result

Throughout this lab session, static routing was replaced with dynamic routing by implementing four distinct protocols: RIP, EIGRP, OSPF, and BGP. Once each protocol was configured, the routers began sharing routing information with one another automatically, making manual route entry unnecessary. Successful configuration was verified by reviewing the routing tables, which reflected newly acquired routes. End-to-end network connectivity was then confirmed through successful ping tests across the entire topology.

Discussion

In this lab, several dynamic routing protocols were configured and tested individually to observe their behavioral differences. RIP proved the simplest to set up, though its reliance on hop count alone limits its effectiveness. OSPF showed greater sophistication by selecting routes based on link cost, which is tied to interface bandwidth. EIGRP exhibited the quickest response to network changes, demonstrating superior convergence speed. BGP illustrated how large-scale inter-network communication functions — by exchanging routing data between autonomous systems

rather than dealing with individual routers. Connectivity across the network was validated using ping tests at the conclusion of each protocol's configuration.

Conclusion

In this laboratory exercise, various dynamic routing algorithms were successfully configured and deployed, demonstrating the practical implementation of automated network path discovery across multiple protocols.